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SENSOR CAP FOR A SENSOR AND METHOD OF MANUFACTURING THE SENSOR CAP

Abstract

A sensor cap for an electrochemical sensor for determining the concentration of an analyte in a measurement medium, includes a main part, which extends along a first axis and has a first end, which has a first opening, and a second end, which is opposite the first end and has a second opening, wherein an electrolyte chamber extends between the first opening and the second opening and a connecting unit is formed at the second end in order to connect the sensor cap to a sensor body of the electrochemical sensor, a membrane unit having a first membrane element and a second membrane element, wherein the first membrane element is permeable to the analyte and the second membrane element comprises a material that is impermeable to the analyte, wherein the second membrane element has a through-hole.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application is related to and claims the priority benefit of German Patent Application No. 10 2024 105 481.6, filed on Feb. 27, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a sensor cap for an electrochemical sensor, a sensor cap for an optochemical sensor, an electrochemical sensor, an optochemical sensor, a manufacturing method of the sensor cap for the electrochemical sensor, and a method of manufacturing the sensor cap for the optochemical sensor.

BACKGROUND

[0003] In analytical measurement technology, for example, in the fields of water management, of environmental analysis, in industry, e.g., in food technology, biotechnology, and pharmaceuticals, as well as for the most varied laboratory applications, measured variables, such as the pH, the conductivity, or even the concentration of analytes, such as ions or dissolved gases in a gaseous or liquid measurement medium, are of great importance. These measured variables can be detected and/or monitored, for example, by means of electrochemical sensors, such as potentiometric, amperometric, voltammetric, coulometric or optochemical sensors.

[0004] Electrochemical and optochemical sensors frequently comprise a membrane, which is brought into contact with a measurement medium, e.g., a measurement gas or a measurement liquid, in order to capture measured values during continuous measurement. Depending on whether the sensor is an electrochemical or optochemical sensor, the membrane serves a different function and is specifically designed for its use. In the case of electrochemical sensors, the membrane is often a porous layer, in particular one with pores or openings, which is at least partially permeable to the measurement medium or at least an analyte contained therein. In the case of optochemical sensors, for example, the membrane is often a thin film, which is used as a carrier material.

[0005] A property of these membranes is usually that the transport of substances through the membrane can take place in both directions, i.e., from the measurement medium to the interior of the sensor and vice versa. This makes it possible, for example, for the analyte to be detected by the sensor to migrate into the interior of the sensor, which is also desirable. However, with electrochemical sensors, for example, which have an electrolyte inside, components of the electrolyte can migrate from the measurement chamber of the sensor through the membrane into the surrounding medium. This results in the service life of the electrolyte being shortened. Consequently, the electrolyte of the sensor must be changed regularly, which increases the maintenance effort of the sensor.

SUMMARY

[0006] It is therefore an object of the present disclosure to provide a sensor that is durable, reliable and accurate.

[0007] This object is achieved according to the present disclosure by a sensor cap for an electrochemical sensor for determining the concentration of an analyte in a measurement medium according to the present disclosure.

[0008] The sensor cap according to the present disclosure comprises: [0009] a main part, which extends along a first axis and has a first end, which has a first opening, and a second end, which is opposite the first end and has a second opening, wherein an electrolyte chamber extends between the first opening and the second opening and a connecting unit is formed at the second end in order to connect the sensor cap to a sensor body of the electrochemical sensor, [0010] a membrane unit having a first membrane element and a second membrane element, [0011] wherein the first membrane element is permeable to the analyte and the second membrane element comprises a material that is impermeable to the analyte, [0012] wherein the second membrane element has a through-hole, wherein the membrane unit is connected to the main part such that the first opening is closed by the membrane unit, [0013] wherein the first membrane element and the second membrane element are welded to the main part.

[0014] The sensor cap according to the present disclosure makes it possible to reduce leaching of the electrolyte arranged in the electrolyte chamber of the sensor head, which increases the service life of the sensor.

[0015] According to one embodiment of the present disclosure, the first membrane element comprises a hydrophilic material, for example, polyethylene terephthalate (PET), and the second membrane element comprises a gas-tight material, for example, polyvinylidene fluoride (PVDF).

[0016] According to a further embodiment of the present disclosure, the main part has a positioning aid in the electrolyte chamber, and a spacer element is arranged between the membrane unit and the positioning aid, wherein the positioning aid holds the spacer element at a predetermined axial position.

[0017] According to one embodiment of the present disclosure, the main part has a pressure equalization opening, which opens into the electrolyte chamber and is closed by a valve unit.

[0018] According to one embodiment of the present disclosure, the main part comprises a plastics material and the membrane unit is ultrasonically welded to the main part.

[0019] The above-mentioned object is also achieved by an electrochemical sensor according to the present disclosure.

[0020] The electrochemical sensor according to the present disclosure has a sensor cap according to the present disclosure and a sensor body having an electrode unit, wherein the electrode unit has an electrode, and wherein the sensor cap is connected to the sensor body via a connecting unit and an electrolyte is arranged in the electrolyte chamber.

[0021] According to one embodiment of the present disclosure, the through-hole of the second membrane element has a predetermined first diameter and the electrode has a predetermined second diameter, wherein the first diameter and the second diameter are substantially identical.

[0022] The above-mentioned object is also achieved by a method for manufacturing a sensor cap according to the present disclosure.

[0023] The method according to the present disclosure comprises: [0024] providing a main part and a membrane unit, wherein the main part extends along a first axis and has a first end, which has a first opening, and a second end, which is opposite the first end and has a second opening, wherein an electrolyte chamber extends between the first opening and the second opening and a connecting unit is formed at the second end in order to connect the sensor cap to a sensor body of the electrochemical sensor, [0025] connecting the membrane unit and the main part so that the first membrane element and the second membrane element are connected to the main part and close the first opening, wherein the first membrane element and the second membrane element are simultaneously ultrasonically welded to the main part.

[0026] According to one embodiment of the present disclosure, the main part has a positioning aid in the electrolyte chamber and, prior to connecting the membrane unit and the main part, a spacer element is inserted into the electrolyte chamber so that the spacer element is held by the positioning aid.

[0027] The above-mentioned object is also achieved by a sensor cap for an optochemical sensor for

determining the concentration of an analyte in a measurement medium according to the present disclosure.

[0028] The sensor cap according to the present disclosure comprises: [0029] a main part, which extends along a first axis and has a first end and a second end opposite the first end, [0030] wherein a window is formed at the first end and a connecting unit is formed at the second end in order to connect the sensor cap to a sensor body of the electrochemical sensor, [0031] a membrane unit having a first membrane element and a second membrane element, [0032] wherein the first membrane element is permeable to the analyte and the second membrane element comprises a material that comprises a luminescent substance, [0033] wherein the membrane unit is connected to the main part such that the window is covered by the membrane unit, [0034] wherein the first membrane element and the second membrane element are welded to the main part.

[0035] The above-mentioned object is also achieved by a method for manufacturing a sensor cap according to the present disclosure.

[0036] The method according to the present disclosure comprises: [0037] providing a main part and a membrane unit, wherein the main part extends along a first axis and has a first end and a second end opposite the first end, wherein a window is formed at the first end, [0038] connecting the membrane unit and the main part so that the first membrane element and the second membrane element are welded to the main part and the window is covered by the membrane unit.

[0039] The above-mentioned object is also achieved by an optochemical sensor according to the present disclosure.

[0040] The optochemical sensor according to the present disclosure has a sensor cap according to the present disclosure and a sensor body, wherein the sensor cap is connected to the sensor body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0041] The present disclosure is explained in more detail with reference to the following figures.

[0042] FIG. 1 shows an embodiment of an electrochemical sensor according to the present disclosure;

[0043] FIG. 2 shows an embodiment of a sensor cap according to the present disclosure for the electrochemical sensor of FIG. 1;

[0044] FIG. 3 shows another embodiment of an optochemical sensor according to the present disclosure; and

[0045] FIG. 4 shows a further embodiment of a sensor cap according to the present disclosure for the optochemical sensor of FIG. 3.

DETAILED DESCRIPTION

[0046] FIG. 1 shows an electrochemical sensor **10** according to the present disclosure having a sensor cap **100** and a sensor body **11**. The sensor body **11** has an electrode unit **12** having an electrode **13**. The electrode unit **12** comprises a counter-electrode **14**. The counter-electrode **14** is formed, for example, as a sleeve. The electrode **13** may be a cathode and the counter-electrode **14** an anode. The sensor cap **100** is detachably connected to the sensor body **11**. The electrochemical sensor **10** is suitable for detecting an analyte **2** in a measurement medium **1**, for example for determining the concentration of the analyte **2** in the measurement medium **1**. The analyte **2**, for example, is free chlorine or bound chlorine.

[0047] FIG. 2 shows the sensor cap **100** in detail in an exploded view. The sensor cap **100** comprises a main part **110**, an electrode **120** and a membrane unit **130**.

[0048] The main part **110** extends along a first axis X and has a first end **111** and a second end **112** opposite the first end **111**. A first opening **113** is formed at the first end **111**. A second opening **114** is formed at the second end **112**. An electrolyte chamber **117** extends between the first opening **113**

and the second opening **114**. The electrolyte chamber **117** is formed, for example, by the entire interior of the main part **110**. The electrolyte chamber **117** can be filled with electrolyte from the second opening **114**.

[0049] The main part **110** comprises plastic, for example PVDF, so that ultrasonic welding to the membrane unit **130** is possible. A positioning aid **115** is preferably formed in the electrolyte chamber **117** in order to position a spacer element **160** relative to an electrode head (not shown) of the electrode **120** in the electrolyte chamber **117**. Preferably, the main part **110** has a plurality of, for example three, positioning aids **115**. The positioning aids **115** are preferably arranged around the first axis X in the main part **110**.

[0050] An electrolyte is stored in the electrolyte chamber **117**.

[0051] The main part **110** preferably has a pressure equalization opening **116**, which opens into the electrolyte chamber **117** and is closed by a valve unit **120**. The valve unit **120** is, for example, an elastic ring or an elastic band, which is stretched around the main part **110** over the pressure equalization opening **116** (see FIG. 2). In order to prevent the elastic band from slipping axially, the main part **110** preferably has an axially circumferential recess that is complementary to the elastic band.

[0052] At the first end **111**, the main part **110** preferably has one or more, in particular three, supports **118**, which preferably extend in parallel with the first axis X so that the electrochemical sensor **10** can be placed on a flat surface with the aid of the three supports **118**, without the membrane unit **130** touching the flat surface. This makes it possible for simple and safe filling of the electrolyte chamber **117** through the second opening **114** when the sensor cap **100** is separated from the sensor body **11**. The support **118** thus allows the sensor cap **100** to be placed stably on a surface.

[0053] At the second end **112**, a connecting unit **119** is formed in order to connect the sensor cap **100** to the sensor body **11** of the electrochemical sensor **10**. The connecting unit **119** is designed, for example, as a thread that is complementary to a thread on the sensor body (not shown).

[0054] The membrane unit **130** is fastened to the main part **110** such that the membrane unit **130** closes the first opening **113**. The membrane unit **130** has a first membrane element **140** and a second membrane element **150**. The membrane unit **130** is welded to the main part **110**.

[0055] The first membrane element **140** is permeable to the analyte **2** and the second membrane element **150** comprises a material that is impermeable to the analyte **2**. The first membrane element **140** is, for example, a sensory hydrophilic membrane. According to an alternative embodiment, it is also possible that the first membrane element **140** has hydrophobic properties, which leads to a lower transport of substances through the first membrane element **140**. The first membrane element **140** comprises a hydrophilic material, for example, PET, and the second membrane element **150** comprises a gas-tight material, for example, PVDF. The first membrane element **140** and the second membrane element **150** preferably have the same diameter. The first membrane element **140** and the second membrane element **150** are connected to the main part **110** by means of ultrasonic welding.

[0056] The second membrane element **150** has a through-hole **151**. The through-hole **151** is preferably arranged centrally in the second membrane element **150**. The through-hole **151** has a predetermined first diameter D1 and the electrode **13** has a predetermined second diameter D2. The first diameter D1 and the second diameter D2 are preferably substantially identical. The electrode **13** is preferably arranged axially aligned with the through-hole **151**, i.e., on the first axis X.

[0057] According to an embodiment not shown, it is also possible to arrange a third membrane element in the membrane unit **130**. The third membrane element is, for example, a support grid or a carrier material with a mediator for an intermediate chemical reaction with the analyte **2**.

[0058] A positioning aid **115** is preferably formed in the electrolyte chamber **117**. In the electrolyte chamber **117**, a spacer element **160** is preferably arranged between the membrane unit **130** and the positioning aid **115**. The positioning aid **115** makes it possible for the spacer element **160** to be held

at a predetermined axial position. The spacer element **160** is, for example, a grid or a porous element. The grid is, for example, made of polyester. The spacer element **160** has a predetermined thickness and is in contact with the membrane unit **130**, preferably with the second membrane element **150**, and with the electrode **13** when the sensor cap **100** is connected to the sensor body **11**, so that the predetermined thickness of the spacer element **160** defines the distance of the electrode **13** to the membrane unit **130**. This makes it possible to ensure a predetermined diffusion path between the electrode **13**, which is preferably used as a cathode, and the membrane unit **130**. This in turn ensures maximum accuracy for the measurements performed with the electrochemical sensor **10**, for example, the so-called slope of the electrochemical sensor **10**.

[0059] The membrane unit **130** is connected to the main part **110** such that the electrolyte chamber **117** is closed by the membrane unit **130**. The second membrane element **150** is arranged between the first membrane element **140** and the main part **110** such that the electrode **13** is arranged at the through-hole **151**. The first membrane element **140** and the second membrane element **150** are jointly connected to the main part **110** by ultrasonic welding. The membrane unit **130** is preferably welded circumferentially at its edge region to the main part **110**. The main part **110** preferably has a so-called concentrator formed at the first end **111** and running about the first axis X (not shown).

[0060] The following describes a method for manufacturing the sensor cap **100** described above.

[0061] Initially, the main part **110** and the membrane unit **130** are provided.

[0062] Next, the membrane unit **130** and the main part **110** are connected so that the first membrane element **140** and the second membrane element **150** are simultaneously connected to the main part **110** and close the first opening **113**. The connection is carried out by ultrasonic welding. When welding the membrane unit **130**, the first membrane element **140** and the second membrane element **150** are welded to the main part **110** at the same time. For this purpose, the first membrane element **140** and the second membrane element **150** are placed on top of each other and jointly welded to the main part **110**. It is advantageous if the first membrane element **140** and the second membrane element **150** have the same diameter.

[0063] Preferably, the membrane unit **130** is welded circumferentially in its edge region.

[0064] Prior to connecting the membrane unit **130** and the main part **110**, the spacer element **160** is preferably inserted onto the positioning aid **115** in the electrolyte chamber **117**.

[0065] In the following, an alternative embodiment of the sensor cap in the application case of an optochemical sensor **20** is described.

[0066] FIG. **3** shows an optochemical sensor **20** having a sensor cap **200** and a sensor body **21** for detecting an analyte **3**, such as oxygen. The sensor cap **200** is detachably connected to the sensor body **21**.

[0067] FIG. **4** shows an enlarged representation of the sensor cap **200**.

[0068] The sensor cap **200** comprises a main part **210** and a membrane unit **230**.

[0069] The main part **210** extends along the first axis X and has a first end **211** and a second end **212** opposite the first end **211**. A window **219** is formed at the first end **211** and a connecting unit **214** is formed at the second end **212** in order to connect the sensor cap **200** to a sensor body **21** of the optochemical sensor **20**. As shown in FIG. **4**, the connecting unit **214** is designed, for example, as an internal thread that is complementary to an external thread (not shown) on the sensor body.

[0070] The membrane unit **230** has a first membrane element **240** and a second membrane element **250**. The first membrane element **240** is permeable to the analyte **3** and at the same time, for example, hydrophilic. The second membrane element **250** comprises a material that comprises a luminescent substance. The second membrane element **250** is also referred to as a functional layer.

[0071] The membrane unit **230** is connected to the main part **210** such that the window is covered by the membrane unit **230**.

[0072] The second membrane element **250** is arranged between the first membrane element **240** and the main part **210**.

[0073] Hereinafter, the method of manufacturing the sensor cap **200** is described.

[0074] Initially, the main part **210** and the membrane unit **230** are provided.
[0075] Then, the membrane unit **230** and the main part **210** are connected so that the first membrane element **240** and the second membrane element **250** are jointly connected to the main part **210** and the window **219** is covered by the membrane unit **230**. The connection is carried out by an ultrasonic welding process, as in the first embodiment described above.

Claims

1. A sensor cap for an electrochemical sensor for determining a concentration of an analyte in a measurement medium, the sensor cap comprising: a main part, which extends along a first axis and has a first end, which includes a first opening, and a second end, which is opposite the first end and includes a second opening, wherein an electrolyte chamber extends between the first opening and the second opening, and wherein a connector is formed at the second end, which is configured to connect the sensor cap to a sensor body of the electrochemical sensor, a membrane unit including a first membrane element and a second membrane element, wherein the first membrane element is permeable to the analyte and the second membrane element comprises a material that is impermeable to the analyte, wherein the second membrane element includes a through-hole, wherein the membrane unit is connected to the main part such that the first opening is closed by the membrane unit, and wherein the first membrane element and the second membrane element are welded to the main part.
2. The sensor cap according to claim 1, wherein the first membrane element comprises a hydrophilic material, and the second membrane element comprises a gas-tight material.
3. The sensor cap according to claim 1, wherein the main part includes a positioning aid in the electrolyte chamber, and a spacer element is arranged between the membrane unit and the positioning aid, wherein the positioning aid holds the spacer element at a predetermined axial position.
4. The sensor cap according to claim 1, wherein the main part includes a pressure equalization opening that opens into the electrolyte chamber and is closed by a valve unit.
5. The sensor cap according to claim 1, wherein the main part comprises a plastics material, and wherein the membrane unit is ultrasonically welded to the main part.
6. An electrochemical sensor comprising a sensor cap according to claim 1, wherein the sensor body includes an electrode unit, wherein the electrode unit includes an electrode, and wherein the sensor cap is connected to the sensor body via the connector, and an electrolyte is disposed in the electrolyte chamber.
7. The electrochemical sensor according to claim 6, wherein the through-hole of the second membrane element has a predetermined first diameter, and the electrode has a predetermined second diameter, wherein the first diameter and the second diameter are substantially identical.
8. A method for manufacturing the sensor cap according to claim 1, the method comprising: providing the main part and the membrane unit; and, connecting the membrane unit and the main part such that the first membrane element and the second membrane element are connected to the main part and close the first opening, wherein the first membrane element and the second membrane element are simultaneously ultrasonically welded to the main part.
9. The method according to claim 8, wherein the main part includes a positioning aid in the electrolyte chamber and, prior to connecting the membrane unit and the main part, a spacer element is inserted into the electrolyte chamber such that the spacer element is held by the positioning aid.
10. A sensor cap for an optochemical sensor for determining a concentration of an analyte in a measurement medium, the sensor cap comprising: a main part, which extends along a first axis and has a first end and a second end opposite the first end, wherein a window is formed at the first end and a connecting unit is formed at the second end in order to connect the sensor cap to a sensor body of the electrochemical sensor, a membrane unit having a first membrane element and a

second membrane element, wherein the first membrane element is permeable to the analyte and the second membrane element comprises a material that comprises a luminescent substance, wherein the membrane unit is connected to the main part such that the window is covered by the membrane unit, wherein the first membrane element and the second membrane element are welded to the main part.

11. A method for manufacturing the sensor cap according to claim 10, the method comprising: providing the main part and the membrane unit, and connecting the membrane unit and the main part such that the first membrane element and the second membrane element are welded to the main part and the window is covered by the membrane unit.

12. An optochemical sensor comprising the sensor cap according to claim 10 and a sensor body, wherein the sensor cap is connected to the sensor body.

13. The sensor cap according to claim 1, wherein the first membrane element comprises PET, and the second membrane element comprises PVDF.
